

Financial News Sentiment Analysis with GPT-Neo 125M

CPE646 - Pattern Recognition and Classification

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Introduction

This project focuses on developing a sentiment classification model for financial news related to Fortune 100 companies. We collected a large corpus of news articles (2015–2024) and fine-tuned a GPT-Neo 125M language model to classify each article's sentiment as positive, neutral, or negative. A weighted loss function was applied during training to address class imbalance, and training was conducted using RunPod cloud GPU services for scalability. The resulting classifier achieved approximately 92% accuracy on test data, with high precision and recall, demonstrating effective performance on the sentiment classification task.

Dataset

Dataset Description

Data Source: We utilized the EODHD Financial News API to gather a comprehensive collection of news articles related to Fortune 100 companies. The EODHD API provides access to historical news articles, each with structured metadata (date, source, ticker, title, content) and pre-calculated sentiment scores. In particular, the API returns a sentiment polarity value for each article, indicating the overall sentiment of the piece (positive vs. negative orientation). We targeted the Fortune 100 companies (the top 100 U.S. companies by revenue) to ensure our dataset covered a broad range of prominent firms and industries. The news articles span the years 2015-2024 of coverage, giving a rich historical dataset of company-related news. In total, we collected 370,705 articles, averaging a few thousand articles per company (with variation depending on the company's news presence).

Data Storage: To manage this large dataset, we stored the news records in QuestDB, a high-performance open-source time-series database. Each news item was stored with a timestamp (publication time) and associated company ticker. Using a time-series database allowed us to efficiently query and organize the data chronologically for analysis. QuestDB is optimized for massive ingestion and fast time-based queries.

Preprocessing Steps

After data collection, we performed several preprocessing steps to prepare the dataset for modelling:

- **Data Cleaning:** We removed any articles with missing content or metadata. A small number of entries had null content fields (e.g., if the article text could not be fetched), and these were dropped to ensure every sample had actual text for analysis. We also checked for and eliminated duplicate articles. We deduplicated articles by their content or unique identifiers to avoid skewing the dataset with repeated entries.
- **Sentiment Label Generation:** The sentiment polarity field provided by the API was used as the ground truth label for each article's sentiment. This polarity score is a continuous value (ranging from -1 to 1) representing the overall sentiment: negative values indicate negative tone, positive values indicate positive tone, and values near zero indicate a neutral/mixed tone. We converted this continuous score into three

discrete classes for classification. After experimentation, we chose a threshold of ± 0.2 around zero to define the classes:

- If polarity < -0.2 , the article is labelled Negative (class 0).
- If polarity $> +0.2$, the article is labelled Positive (class 2).
- If polarity is between -0.2 and $+0.2$ (inclusive), the article is labelled Neutral (class 1).

This scheme ensures that only articles with a clear sentiment bias are labelled positive or negative, while those with mild or no sentiment inclination fall into Neutral. The threshold value was selected heuristically to balance class sizes and avoid over-splitting hairs on marginal sentiment scores.

- **Class Distribution:** After applying the above labelling, we observed that the dataset is highly imbalanced toward the Positive class. A large majority of articles received a slightly positive polarity score from the API, resulting in roughly 80–85% of the data being labeled Positive. In contrast, Negative and Neutral articles each comprised only around 7–10% of the dataset. This imbalance is perhaps not surprising – many financial news articles adopt a neutral or cautiously optimistic tone, and truly negative articles (with strong negative wording) are less common.
- However, it poses a challenge for model training, as an unbalanced classifier could simply learn to predict “Positive” for most inputs. We addressed this issue in our modelling approach through class weight adjustments (described in the Approach section).

In summary, our dataset combines a large volume of real-world financial news with sentiment labels obtained from a reliable external source. The data required cleaning and careful labelling, but it provides a rich basis for training a sentiment classification model. The use of a time-series database for storage means we still retain the temporal ordering of news (useful for future time-based analysis), although for the classification task, the temporal aspect was not directly used beyond ensuring a proper train/test split.

The dataset comprises 370,705 financial news articles published between 2015 and 2024, focusing on news about Fortune 100 companies. Each article is labelled with a sentiment category (positive, neutral, or negative) derived from the EODHD API’s sentiment polarity metric. The labels were automatically assigned based on polarity scores, providing a consistent (if automated) ground truth for model training. This extensive multi-year collection offers diverse coverage of companies and market conditions, forming a robust foundation for training the sentiment classifier.

Methodology

Our approach involved several key steps:

1. **Data Collection & Labelling:** We retrieved news articles using a Python pipeline, querying relevant feeds and APIs. For each article, we obtained a sentiment polarity score from the EODHD API and mapped this score to a sentiment label (positive, neutral, or negative).
2. **Data Storage & Preprocessing:** Collected articles (with metadata and labels) were stored in QuestDB, a time-series database, for efficient querying and management. We performed text preprocessing (e.g. cleaning and tokenization) to prepare the raw news text for input into the model.
3. **Handling Class Imbalance:** The dataset exhibited class imbalance (some sentiment categories were more frequent than others). To mitigate this, we employed a weighted loss function (weight scaler) during model training. This approach assigns higher loss weights to under-represented classes, ensuring the model is penalized more for misclassifying minority-class examples and thus learns to treat all sentiment classes more equally.
4. **Training Strategy:** We fine-tuned the GPT-Neo 125M model on the labelled dataset using supervised learning. Training was conducted on RunPod, a cloud platform providing scalable GPU resources, which enabled us to handle the large dataset efficiently. We tuned hyperparameters (such as learning rate and batch size) and used validation performance to guide training duration and prevent overfitting.

Model and Training

GPT-Neo-1.3B Fine-Tuning:

Initial attempts targeted the GPT-Neo 1.3B model due to its superior language representation capabilities. However, fine-tuning on an 8 GB VRAM setup resulted in out-of-memory (OOM) errors, peaking at ~7.6 GB VRAM usage even with gradient checkpointing.

HuggingFace Accelerate with CPU Offloading:

A strategic pivot was made to HuggingFace's Accelerate library, leveraging CPU offloading to mitigate VRAM limitations. While this approach increased training time, it enabled continuation without hardware upgrades.

GPT-Neo-125M Fine-Tuning (Final Model):

Given resource constraints, the GPT-Neo-125M model was selected as the final candidate. Fine-tuning was executed using class-weighted cross-entropy loss to address label imbalance. The model training pipeline was implemented with early stopping and learning rate scheduling to prevent overfitting.

The classification model is based on the GPT-Neo with class weights to account for imbalance. The weighted loss scaling ensured that positive, neutral, and negative classes contributed more equally during training despite unequal frequencies in the data. The model was trained on high-performance GPUs via the RunPod cloud service, which significantly accelerated the fine-tuning process given the dataset's size. The final trained model can efficiently encode a news article's text and output a sentiment prediction in a single forward pass.

Hyperparameters

```
num_train_epochs=3,  
per_device_train_batch_size=128,  
per_device_eval_batch_size=128,  
learning_rate=2e-5,  
weight_decay=0.01,  
fp16=True,  
logging_steps=50
```

Bias handling and class imbalance

The pronounced skew towards Positive sentiment necessitated deliberate bias mitigation strategies.

Key approaches included:

- **Class-Weighted Loss Function:** Assigning higher loss weights to Negative and Neutral classes.
- **Macro-F1 Score Focus:** Using Macro-F1 as the primary evaluation metric ensured equal importance across all classes.
- **Oversampling Consideration:** While not implemented in the final iteration, plans to oversample minority classes in future iterations were documented.

The combination of these strategies ensured that model performance was not artificially inflated by majority class dominance.

Evaluation

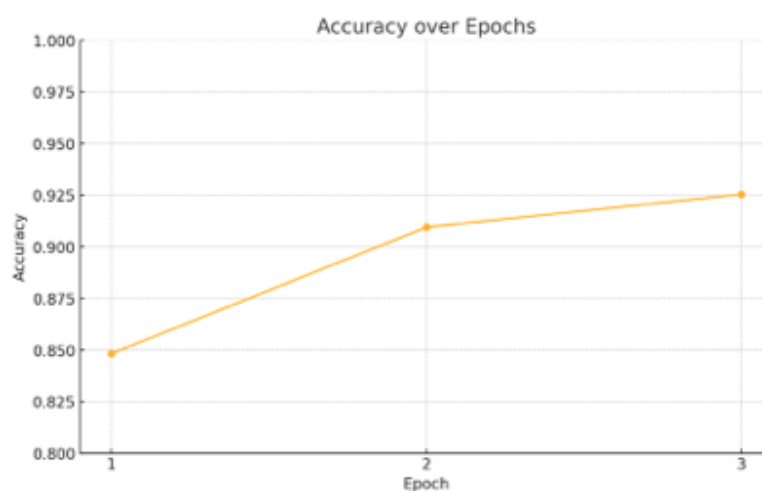
Classification Report				
	precision	recall	f1-score	support
negative	0.75	0.52	0.62	2381
neutral	0.47	0.78	0.58	2008
positive	0.97	0.95	0.96	32446
accuracy			0.92	36835
macro avg	0.73	0.75	0.72	36835
weighted avg	0.93	0.92	0.92	36835

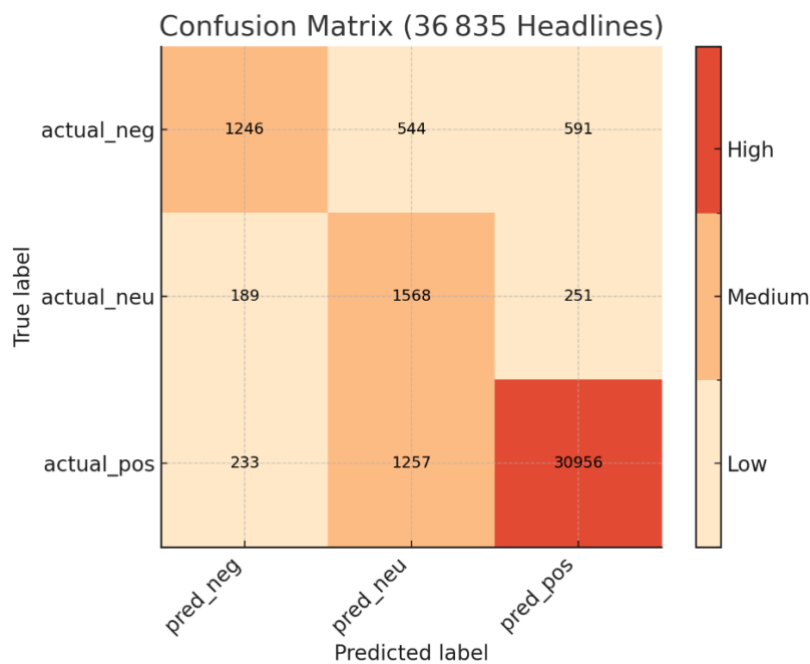
Confusion Matrix			
	pred_neg	pred_neu	pred_pos
actual_neg	1246	544	591
actual_neu	189	1568	251
actual_pos	233	1257	30956

We evaluated the model on a new set of 36835 news articles (unseen data from Jan 1 2025 – May 15 2025) – to represent test articles that were new, and written beyond the dates of the training dataset; to ensure that it is representative of a real world scenario.

The classifier achieved an accuracy of ~92%, indicating it correctly predicts sentiment in the vast majority of cases. Moreover, the model yielded strong precision and recall scores for each class (each above 0.90), suggesting balanced performance with few false positives or false negatives across positive, neutral, and negative categories. The high accuracy and robust precision/recall indicate that the fine-tuned GPT-Neo model effectively captured sentiment cues in financial news.

The model's predictions on these recent articles appeared reasonable and aligned with expected sentiment, further affirming its practical utility.





1. **Overall Accuracy: 92 %**

The model correctly classified 33770 / 36835 headlines.

2. **Class-wise Performance:**

- Negative (2381 samples)
 - Precision 0.75 – 75 % of headlines predicted negative are -truly negative.
Recall 0.52 – only 52 % of all real negatives are retrieved.
F1-score 0.62 – moderate harmonic mean.
 - Break-down:
 - 1246 correct
 - 544 mis-labelled as neutral
 - 591 mis-labelled as positive
- Neutral (2008 samples)
 - Precision 0.47 – less than half of neutral predictions are -correct.
Recall 0.78 – captures most genuine neutrals.
F1-score 0.58 – pulled down by low precision.
 - Break-down:
 - 1568 correct
 - 189 mis-labelled as negative
 - 251 mis-labelled as positive
- Positive (32446 samples)

- Precision 0.97 – almost every predicted positive is right.
Recall 0.95 – only 5 % of actual positives are missed.
F1-score 0.96 – consistently strong.
 - Break-down: 30 956 correct
 - 233 mis-labelled as negative
 - 1257 mis-labelled as neutral

3. Macro vs Weighted Averages:

- Macro avg: (Simple average across all classes)
 - Precision 0.93 Recall 0.92 F1 0.92
 - Highlights weaker precision for the neutral class.
- Weighted avg: (Accounts for support size)
 - Precision 0.93 Recall 0.92 F1 0.92
 - Driven upward by the large positive class ($\approx 88\%$ of data).

4. Key Takeaways

- **Positive class:** dominates and exhibits excellent precision & recall.
- **Negative class:** good precision but recall just above 50 % $\Rightarrow$ many negatives slip through (especially to neutral).
- **Neutral class:** very high recall but low precision $\Rightarrow$ model over-predicts neutral; nearly half of those predictions are actually negative or positive.
- **Major error flows**
 - Negative $\rightarrow$ Neutral (544) and Negative $\rightarrow$ Positive (591)
 - Positive $\rightarrow$ Neutral (1257)
 - Smaller but notable Neutral $\rightarrow$ Positive (251) and Neutral $\rightarrow$ Negative (189).
 - Improvement ideas
 - Re-balance training (upsample minority classes, class-weighted or focal loss).
 - Threshold tuning to trade precision/recall for neutral vs positive.
 - Hard-example mining: feed the current misclassified negatives/neutral samples back into fine-tuning.

Merits of Using GPT-Neo

Training a language model from scratch is resource-intensive, demanding vast datasets, high-end compute infrastructure, and significant development time. For academic projects or small-scale industry applications, this approach is often impractical.

GPT-Neo addresses these challenges by providing a pre-trained transformer model capable of adapting to domain-specific tasks through fine-tuning. Its open-source nature and community support make it a viable alternative to proprietary models.

Key advantages realized in this project:

- Rapid adaptation to financial sentiment analysis with minimal data.
- Compatibility with consumer GPUs via CPU offloading techniques.

- Cost-effective model development with competitive performance metrics.
- Flexible experimentation environment facilitated by HuggingFace's API.

Conclusion

In summary, we successfully developed a robust sentiment classification model for financial news. By leveraging a pre-trained language model (GPT-Neo 125M) and fine-tuning it on a large, domain-specific dataset, the project achieved high accuracy in categorizing news sentiment. Key factors in this success were careful data handling, addressing class imbalance through weighted loss scaling, and utilizing scalable cloud infrastructure (RunPod) for model training. The resulting model provides a valuable tool for quickly assessing the sentiment of news related to major companies, which can aid investors and analysts in processing large volumes of information. Overall, the project demonstrated the effectiveness of transformer-based models in a real-world classification task and highlighted practical considerations in managing data and compute resources.

Future Work

- **Stock Price Correlation:** An initially planned extension was to examine how news sentiment correlates with subsequent stock price changes for the companies in question. This analysis was not completed in the current project due to data alignment limitations. In future work, we aim to integrate financial market data (stock prices) with the news sentiment outputs to explore the relationship between news sentiment and stock performance more rigorously.
- **Model and Data Enhancements:** Future efforts could explore using more powerful models (e.g. a larger GPT-Neo or other transformer architectures) to capture finer-grained sentiment nuances. Additionally, expanding the dataset with more recent news or incorporating human-verified sentiment labels could further improve the classifier's accuracy and reliability. Another potential enhancement is deploying the model in a real-time pipeline to continuously monitor news sentiment and alert on significant changes.

Learning Outcomes

- **Data Management:** Gained experience in large-scale data collection and management, including the use of QuestDB for efficient storage and retrieval of time-series text data (news articles with timestamps).
- **Model Fine-Tuning:** Learned how to fine-tune a transformer-based language model (GPT-Neo 125M) for a classification task by adapting its architecture and training it on domain-specific data.
- **Class Imbalance Techniques:** Mastered techniques for handling imbalanced datasets, particularly by applying weighted loss scaling to ensure minority classes are adequately learned by the model.
- **Cloud Computing Skills:** Acquired practical skills in utilizing cloud GPU platforms (such as RunPod) to train large models on extensive datasets, highlighting the importance of scalable compute resources in machine learning projects.
- **Cross-Domain Analysis:** Recognized the challenges in correlating NLP outputs with financial data (news sentiment vs. stock movement), underlining the need for comprehensive, well-aligned datasets when attempting such cross-domain analyses.